Per ELIZA



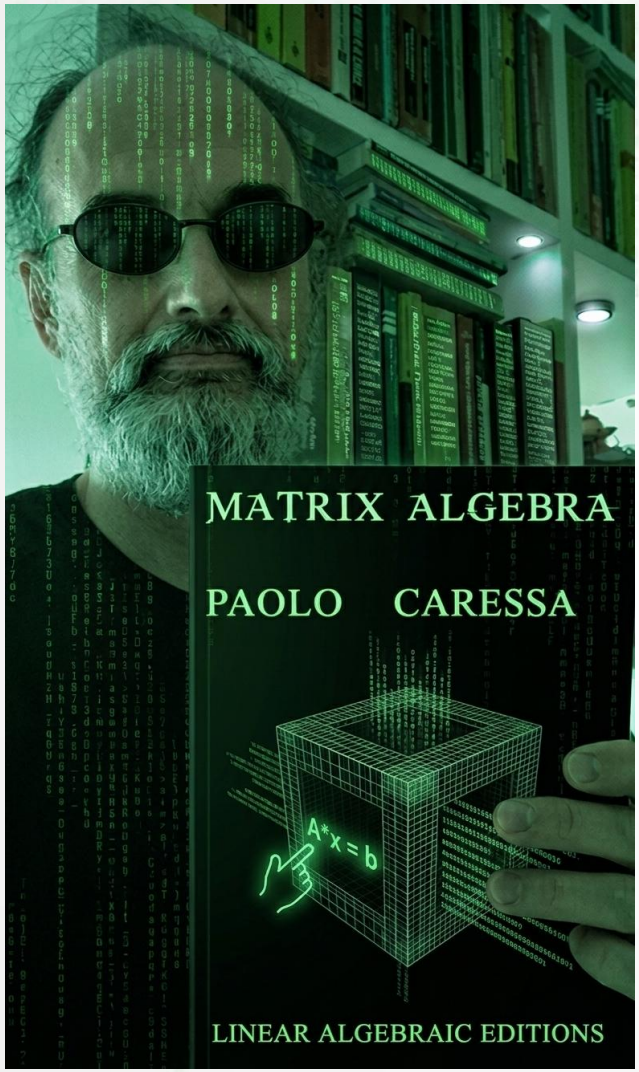
Un chatbot AI di 60 anni fa



Paolo Caressa

IT Expert @ GSE spa — Extern Researcher @ M&MoCS

Pescara DevFest, 7 novembre 2026



Paolo Caressa

1980s: Programmazione BASIC, Forth, Assembly Z80, Pascal.

1990s: Laurea e dottorato in matematica pura (algebra e geometria).

2000s: Ricerca & sviluppo (anche di programmi).

2010s: Project Manager & Professore Aggiunto.

2020s: IT Expert. Continuo a programmare (C, JS, Python) e a interessarmi di AI.
Scrivo, parlo e faccio video.



GLI ANTENATI DI ELIZA



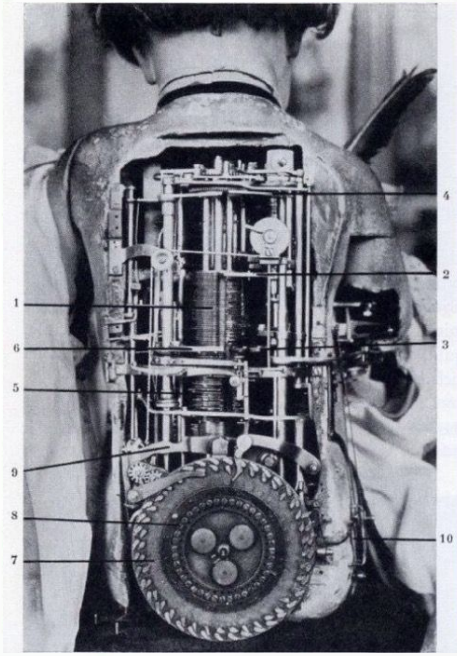
La smania di replicare noi stessi è antica quanto l’umanità



Henri Maillardet (1745-1830)



Pierre Jaquet-Droz (1721-1790)





Wolfgang von Kempelen (1734-1804)



Wolfgangs von Kempelen
k. k. wirklichen Hofraths

Mechanismus

der

menschlichen Sprache

nebst der Beschreibung

seiner sprechenden

Maschine.

Mit XXVII Kupfertafeln.

Hasce igitur penitus voces cum corpore nostro
Exprimimus, rectoque foras emittimus ore,
Mobilis articulat verborum dædala lingua,
Formaturaque labrorum pro parte figurat.
Lucret. lib. IV. v. 553.

Wien,
bei J. V. Degen, 1791.

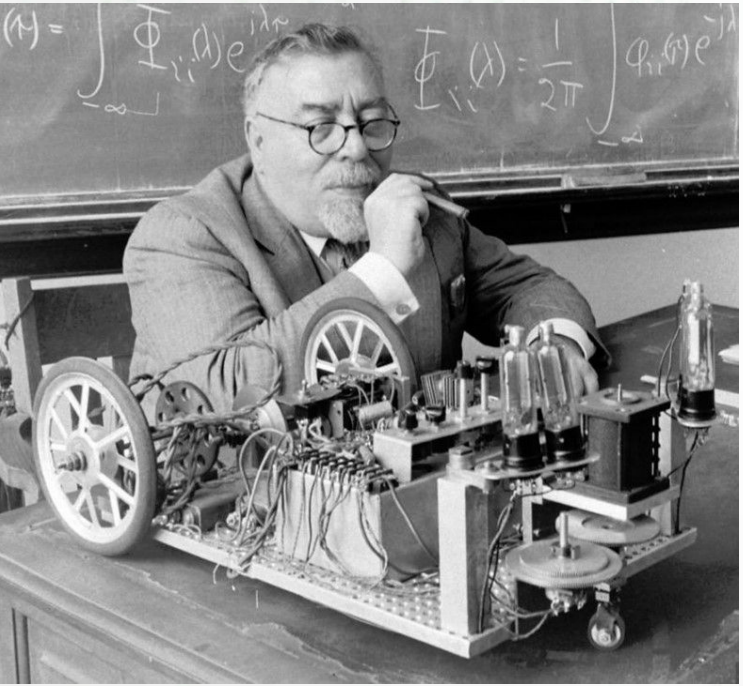


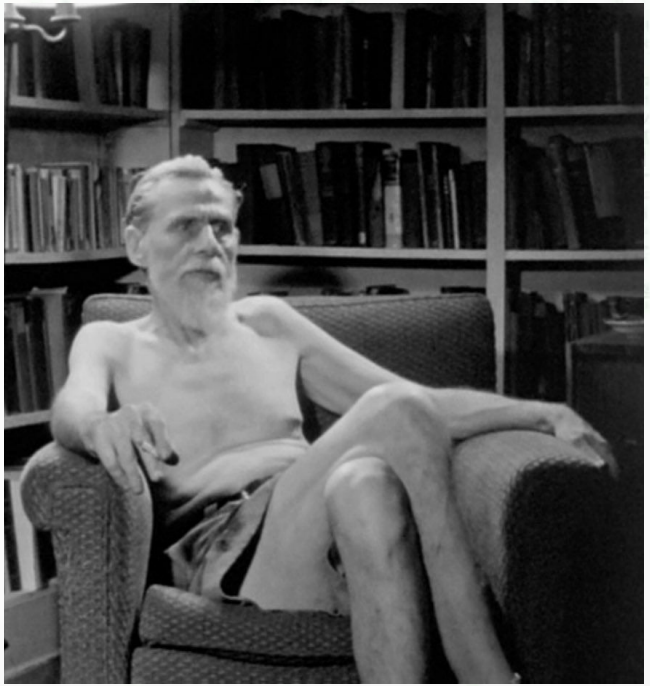


LA NASCITA DELL’AI



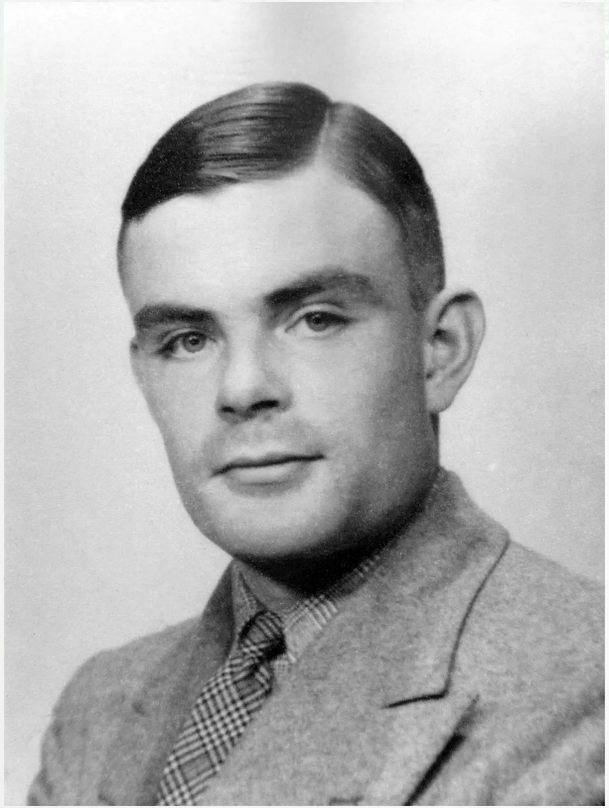
Negli anni ‘40 nasce la cibernetica







Con Alan Turing nasce l'Intelligent Machinery (e molto altro)



1936: Definisce la nozione di algoritmo

1940-45: Decifra i codici nazisti.

1947: Progetta il primo computer UK

1948: Fonda l'intelligenza artificiale (ML, reti neurali etc.)

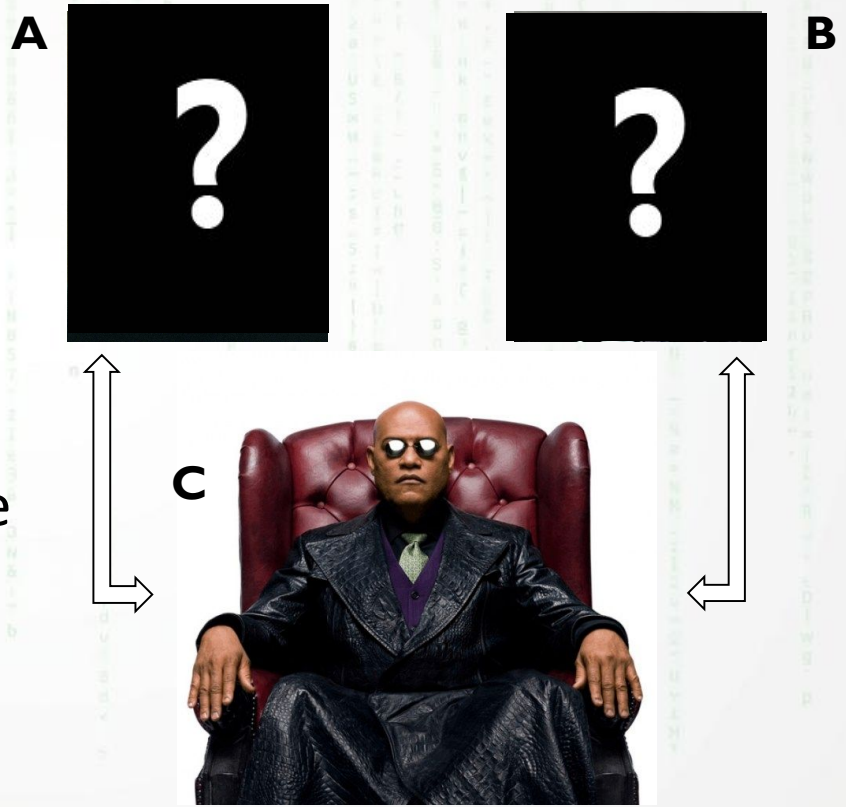
1950: Introduce il test di Turing

1952: Studia fenomeni biologici emergenti

Test di Turing, *Computing machines and intelligence* (1950)

Turing non dà una definizione astratta di intelligenza, ma "operativa": il *gioco dell'imitazione*.

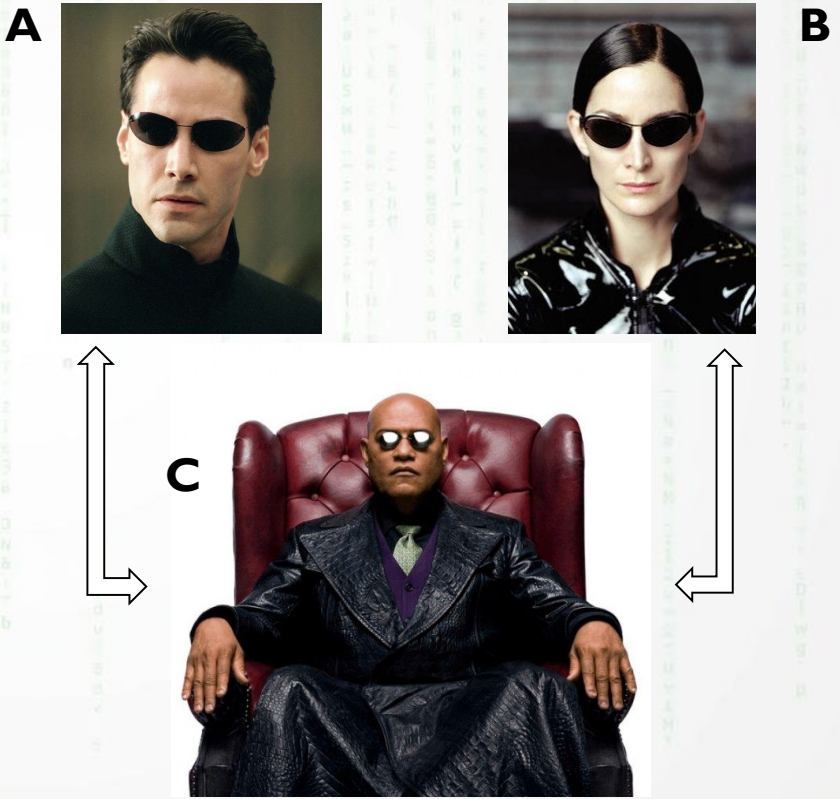
Un computer che riesca a vincere questo gioco è, secondo Turing, intelligente.



Test di Turing, *Computing machines and intelligence* (1950)

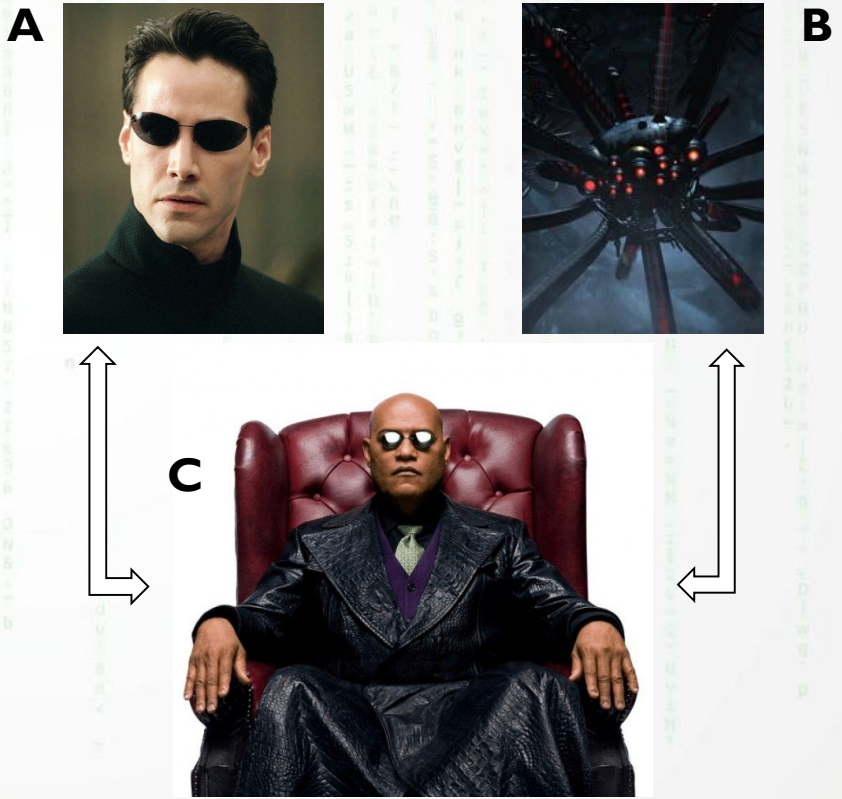
A e B sono di sessi opposti, il sesso di C è ininfluente. L'inquirente C deve capire, facendo domande e ricevendo risposte, se A è l'uomo e B la donna o viceversa.

C e B vincono se C indovina, A vince se C sbaglia.



Test di Turing, *Computing machines and intelligence* (1950)

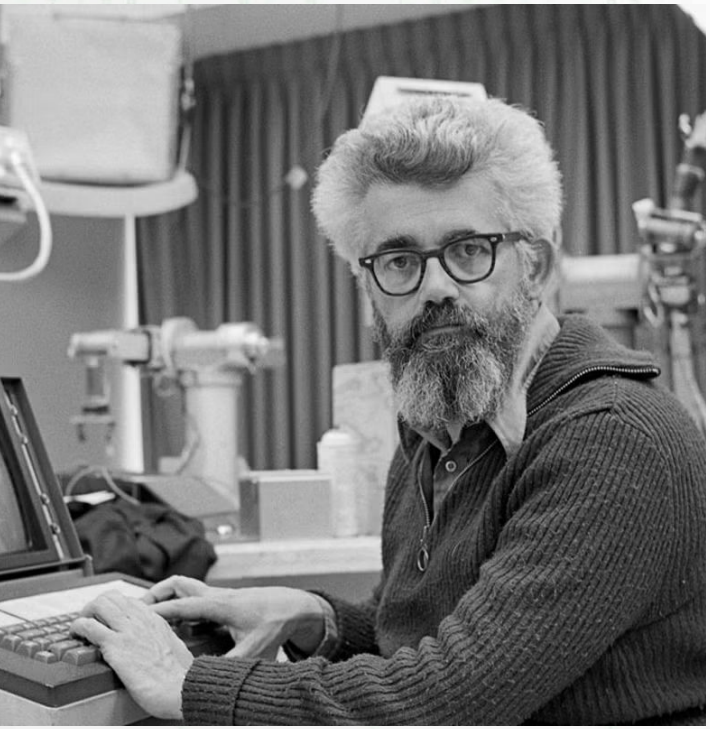
Per Turing una macchina è intelligente se riesce a giocare il gioco dell'imitazione al posto della donna, vincendolo contro i giocatori in carne e ossa in un numero rilevante di casi (70% in su) dopo aver giocato un numero fisso di minuti, per esempio 5.



Reload: Intelligenza Artificiale (Dartmouth College, 1956)



John McCarthy (1927-2011)



1956: Conia il termine Intelligenza Artificiale.

1958: Inventa il Lisp.

1958: Propone di introdurre ricorsività e operatori condizionali nei linguaggi.

1959: Inventa la garbage collection.

1960: Applica il lambda-calcolo allo studio dei linguaggi di programmazione.

1961: Contributo al Time-Sharing

Le idee di base dell'AI simbolica

Rappresentare il problema tramite strutture di dati.

Rappresentare lo spazio delle soluzioni del problema come un grafo da navigare per trovare la soluzione ottimale.

Usare tecniche di logica matematica e teoria combinatoria.

Usare euristiche mediate dalla teoria dei giochi e dalla teoria dell'ottimizzazione combinatoria.

Usare il pattern matching e regole di riscrittura.



ELIZA



Joseph Weizenbaum (1923-2008)



1956: Lavora al sistema bancario ERMA.

1962: Scrive un breve articolo *How to Make a Computer Appear Intelligent* su Datamation.

1963: Propone Slip, un linguaggio analogo al Lisp ma “embeddabile” nel Fortran.

1966: Progetta e programma ELIZA, un chatbot conversazionale, scritto in SLIP.

1976: Pubblica il libro *Computer Power and Human Reason* dove esprime preoccupazione per la diffusione degli strumenti IT.

ELIZA



Il nome “ELIZA” viene dalla protagonista della commedia di George Bernard Shaw *Pigmalione*, ispirata al mito classico dell’artefice di una statua di cui si innamora e che poi prende vita.

Nella commedia un professore di fonetica cerca di plasmare una popolana per farne una dama colta.

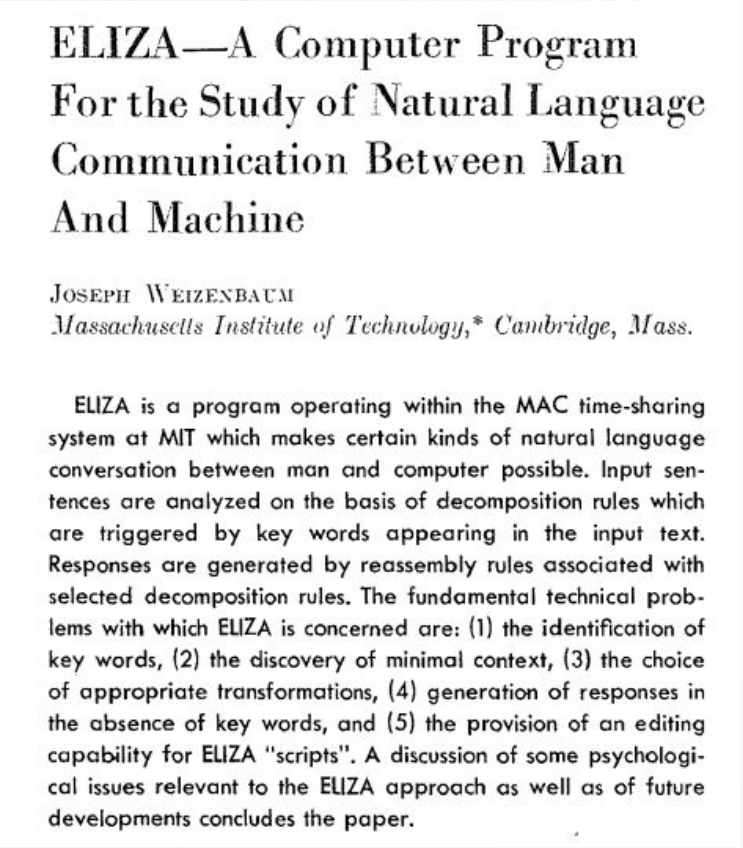
Come è stato possibile programmare un chatbot nel 1966?



l'IBM 7094 era un computer con 32K parole di 36 bit di RAM, 36Mb di disco rigido, unità a nastro.

Al MIT ne avevano due sui quali girava CTSS, il "nonno" di Unix

Idea fondamentale: separare algoritmo e dati



ELIZA—A Computer Program For the Study of Natural Language Communication Between Man And Machine

JOSEPH WEIZENBAUM
Massachusetts Institute of Technology, Cambridge, Mass.*

ELIZA is a program operating within the MAC time-sharing system at MIT which makes certain kinds of natural language conversation between man and computer possible. Input sentences are analyzed on the basis of decomposition rules which are triggered by key words appearing in the input text. Responses are generated by reassembly rules associated with selected decomposition rules. The fundamental technical problems with which ELIZA is concerned are: (1) the identification of key words, (2) the discovery of minimal context, (3) the choice of appropriate transformations, (4) generation of responses in the absence of key words, and (5) the provision of an editing capability for ELIZA "scripts". A discussion of some psychological issues relevant to the ELIZA approach as well as of future developments concludes the paper.

ELIZA è un “motore” di pattern matching che utilizza un insieme di regole, descritte da S-espressioni (il formato dati del Lisp: oggi sarebbe un Json), che viene eseguito a runtime dal motore.

Insomma un interprete per un linguaggio di descrizione di dati, come sarà dieci anni dopo il Prolog, per esempio.

ELIZA è una Gen AI

Durante la chat con l'utente, ELIZA trasforma la domanda dell'utente in una risposta applicando delle "regole di riscrittura", un *leitmotiv* dell'AI simbolica (con una forte teoria logica dietro: lambda calcolo, Church-Rosser etc.).

A ben vedere, sebbene fatto con altri mezzi, questa elaborazione è analoga alla produzione di un output da parte di una Gen AI a partire da un prompt.

Analogia con linguaggi interpretati e shell

ELIZA, per poter trasformare il proprio input nel proprio output, utilizza un insieme di *regole di riscrittura*, che possono essere viste come un programma.

Weizenbaum chiama *script* questo programma dichiarativo di descrizione di dati, che usa la sintassi del Lisp (S-espressioni).

Lo script viene convertito in un formato interno (doubly linked list) e poi eseguito da ELIZA durante la sessione interattiva, che è quindi un vero e proprio REPL.

Il meccanismo di attenzione di ELIZA

Ogni regola è associata a una “parola chiave”, che è quella su cui ELIZA pone la sua attenzione quando la incontra. Se ne incontra più di una, sceglie quella che ha “rango” massimo, un valore che ne indica la precedenza rispetto alle altre (default = 0).

Una volta scelta la parola chiave della domanda dell’utente, ELIZA cerca fra le regole associate alla parola chiave quella il cui pattern ha un match con la frase dell’utente e, se trova un match, applica la sostituzione e risponde, altrimenti dà una risposta generica.

Esempio di regola (preso dallo script DOCTOR)

```
(YOUR = MY
     ((0 MY 0)
      (WHY ARE YOU CONCERNED OVER MY 3)
      (WHAT ABOUT YOUR OWN 3)
      (ARE YOU WORRIED ABOUT SOMEONE ELSE'S 3)
      (REALLY, MY 3)))
```

?WHAT'S YOUR NAME
=> WHY ARE YOU CONCERNED OVER MY NAME
WHAT ABOUT YOUR OWN NAME
ARE YOU REALLY WORRIED ABOUT SOMEONE ELSE'S NAME
REALLY, MY NAME



La mia implementazione

NO AI WERE HARMED DURING THE PREPARATION OF THIS TALK!!!



Un esercizio di anacronismo

ELIZA è uno dei programmi più reimplementati della storia, spesso male, senza seguire le indicazioni di Weizenbaum (io stesso, 40 anni fa, ne predisposi una versione per ZX Spectrum che tentava di parlare italiano).

Negli anni ‘60 fu portato in Lisp e in Snobol, l’unico linguaggio, oltre a Comit, che all’epoca consentiva nativamente il pattern matching.

Un po’ di sano divertimento

Per questo talk, ho implementato ELIZA in Basic delle origini, il Dartmouth Basic, non lo street Basic dei microcomputer anni ‘80.

Ho scritto in vanilla JS un emulatore di terminale Basic per la versione del linguaggio del 1967, la prima a trattare le stringhe.

Poi in questo Basic primitivo ho scritto un ELIZA seguendo le indicazioni di Weizenbaum e ho progettato il formato dei dati leggibile dal Basic dell’epoca.

WAKE UP, USER... THE SYSTEM HAS YOU.

```
YOUR FATHER
?YES, HE NEVER BOTHER ABOUT ME
YOU SEEM QUITE POSITIVE
?NOT AT ALL, MY FATHER HATES ME
I AM NOT SURE I UNDERSTAND YOU FULLY
?MY FATHER HATES ME
TELL ME MORE ABOUT YOUR FAMILY
?MY FATHER IS VIOLENT
WHO ELSE IN YOUR FAMILY IS VIOLENT
?MY BROTHER
WHY DO YOU SAY YOUR BROTHER
?BECAUSE HE BEAT ME
IS THAT THE REAL REASON
?I SUPPOSE. I SUFFER TOO MUCH FOR ALL THAT
YOU SAY YOU SUFFER TOO MUCH FOR ALL THAT
?LIKE AN ANIMAL
PLEASE GO ON
?I HAVE ONLY ONE REASON TO LIVE
CAN YOU ELABORATE ON THAT
?I LIKE TO WRITE CODE
DO YOU SAY YOU LIKE TO WRITE CODE  FOR SOME SPECIAL REASON
?BECAUSE I LOVE PROGRAMMING
THAT'S QUITE INTERESTING
?I ONLY FEEL GOOD IN FRONT OF SOME OLD C CODE
?
```



Perché ELIZA è importante



Perché sembrava che funzionasse!

A typical conversation is the following:

Men are all alike.
IN WHAT WAY
They're always bugging us about something or other.
CAN YOU THINK OF A SPECIFIC EXAMPLE
Well, my boyfriend made me come here.
YOUR BOYFRIEND MADE YOU COME HERE
He says I'm depressed much of the time.
I AM SORRY TO HEAR YOU ARE DEPRESSED
It's true. I am unhappy.
DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY
I need some help, that much seems certain.
WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP
Perhaps I could learn to get along with my mother.
TELL ME MORE ABOUT YOUR FAMILY
My mother takes care of me.
WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU
My father.
YOUR FATHER
You are like my father in some ways.
WHAT RESEMBLANCE DO YOU SEE
You are not very aggressive but I think you don't want me to notice that.
WHAT MAKES YOU THINK I AM NOT VERY AGGRESSIVE
You don't argue with me.
WHY DO YOU THINK I DON'T ARGUE WITH YOU
You are afraid of me.
DOES IT PLEASE YOU TO BELIEVE I AM AFRAID OF YOU
My father is afraid of everybody.
WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER
Bullies.
DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR BOYFRIEND MADE YOU COME HERE

Perché è stata la prima AI a passare il test di Turing

With ELIZA as the basic vehicle, experiments may be set up in which the subjects find it credible to believe that the responses which appear on his typewriter are generated by a human sitting at a similar instrument in another room. How must the script be written in order to maintain the credibility of this idea over a long period of time? How can the performance of ELIZA be systematically degraded in order to achieve controlled and predictable thresholds of credibility in the subject? What, in all this, is the role of the initial instruction to the subject? On the other hand, suppose the subject is told he is communicating with a machine. What is he led to believe about the machine as a result of his conversational experience with it? Some subjects have been very hard to convince that ELIZA (with its present script) is *not* human. This is a striking form of Turing's test. What experimental design would make it more nearly rigorous and airtight?

Perché ha innescato domande su come rapportarsi alle AI!

The whole issue of the credibility (to humans) of machine output demands investigation. Important decisions increasingly tend to be made in response to computer output. The ultimately responsible human interpreter of "What the machine says" is, not unlike the correspondent with ELIZA, constantly faced with the need to make credibility judgments. ELIZA shows, if nothing else, how easy it is to create and maintain the illusion of understanding, hence perhaps of judgment

Perché ha fatto la storia del NLP prima di ChatGPT!

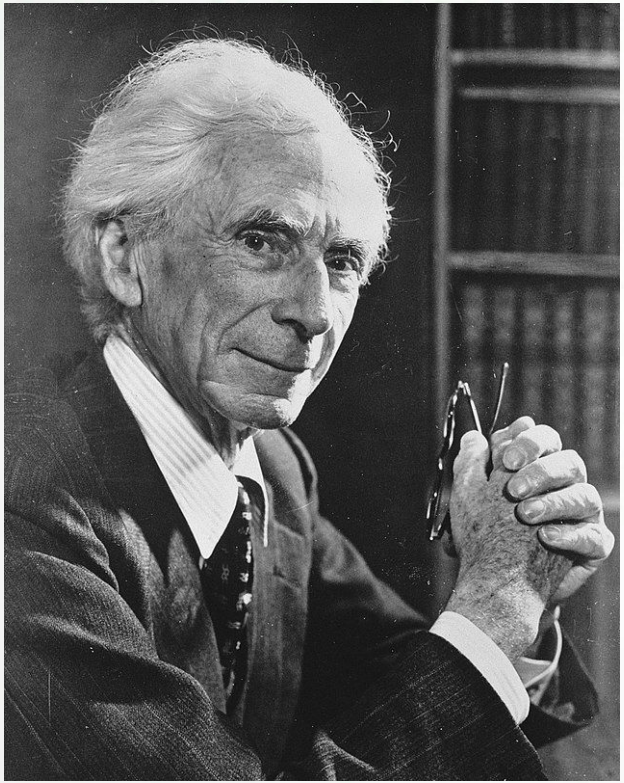
One goal for an augmented ELIZA program is thus a system which already has access to a store of information about some aspects of the real world and which, by means of conversational interaction with people, can reveal both what it knows, i.e., behave as an information retrieval system, and where its knowledge ends and needs to be augmented. Hopefully the augmentation of its knowledge will also be a direct consequence of its conversational experience. It is precisely the prospect that such a program will converse with many people and learn something from each of them, which leads to the hope that it will prove an interesting and even useful conversational partner.



ALLORA, PERCHÉ l’AI SIMBOLICA HA FALLITO?



Perché i matematici ci avevano già provato







Perché ha ignorato tutte le idee di Turing

If we are trying to produce an intelligent machine, and are following the human model as closely as we can, we should begin with a machine with very little capacity to carry out elaborate operations or to react in a disciplined manner to orders (taking the form of interference). Then by applying appropriate interference, mimicking education, we should hope to modify the machine until it could be relied on to produce definite reactions to certain commands. This would be the beginning of the process. I will not attempt to follow it further now.

Intelligent Machinery (1948)

Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child brain is something like a notebook as one buys it from the stationer's. Rather little mechanism, and lots of blank sheets. (Mechanism and writing are from our point of view almost synonymous.) Our hope is that there is so little mechanism in the child brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

Computing machinery & intelligence (1950)

Perché ha ignorato tutte le idee di Turing

It is probably wise to include a random element in a learning machine. A random element is rather useful when we are searching for a solution of some problem. Suppose for instance we wanted to find a number between 50 and 200 which was equal to the square of the sum of its digits, we might start at 51 then try 52 and go on until we got a number that worked. Alternatively we might choose numbers at random until we got a good one. This method has the advantage that it is unnecessary to keep track of the values that have been tried, but the disadvantage that one may try the same one twice, but this is not very important if there are several solutions. The systematic method has the disadvantage that there may be an enormous block without any solutions in the region which has to be investigated first, Now the learning process may be regarded as a search for a form of behaviour which will satisfy the teacher (or some other criterion). Since there is probably a very large number of satisfactory solutions the random method seems to be better than the systematic. It should be noticed that it is used in the analogous process of evolution. But there the systematic method is not possible. How could one keep track of the different genetical combinations that had been tried, so as to avoid trying them again?

Computing machinery & intelligence (1950)

Perché ha confuso architettura e funzionalità



Intelligence without representation*

Rodney A. Brooks

MIT Artificial Intelligence Laboratory, 545 Technology Square, Rm. 836, Cambridge, MA 02139, USA

Received September 1987

Brooks, R.A., Intelligence without representation, Artificial Intelligence 47 (1991), 139–159.

* This report describes research done at the Artificial Intelligence Laboratory of the Massachusetts Institute of Technology. Support for the research is provided in part by an IBM Faculty 9 Development Award, in part by a grant from the Systems Development Foundation, in part by the University Research Initiative under Office of Naval Research contract N00014-86-K-0685 and in part by the Advanced Research Projects Agency under Office of Naval Research contract N00014-85-K-0124.

Abstract

Artificial intelligence research has foundered on the issue of representation. When intelligence is approached in an incremental manner, with strict reliance on interfacing to the real world through perception and action, reliance on representation disappears. In this paper we outline our approach to incrementally building complete intelligent Creatures. The fundamental decomposition of the intelligent system is not into independent information processing units which must interface with each other via representations. Instead, the intelligent system is decomposed into independent and parallel activity producers which all interface directly to the world through perception and action, rather than interface to each other particularly much. The notions of central and peripheral systems evaporateeverything is both central and peripheral. Based on these principles we have built a very successful series of mobile robots which operate without supervision as Creatures in standard office environments.

Perché i computer sono fatti per fare calcoli

BULLETIN OF
MATHEMATICAL BIOPHYSICS
VOLUME 5, 1943

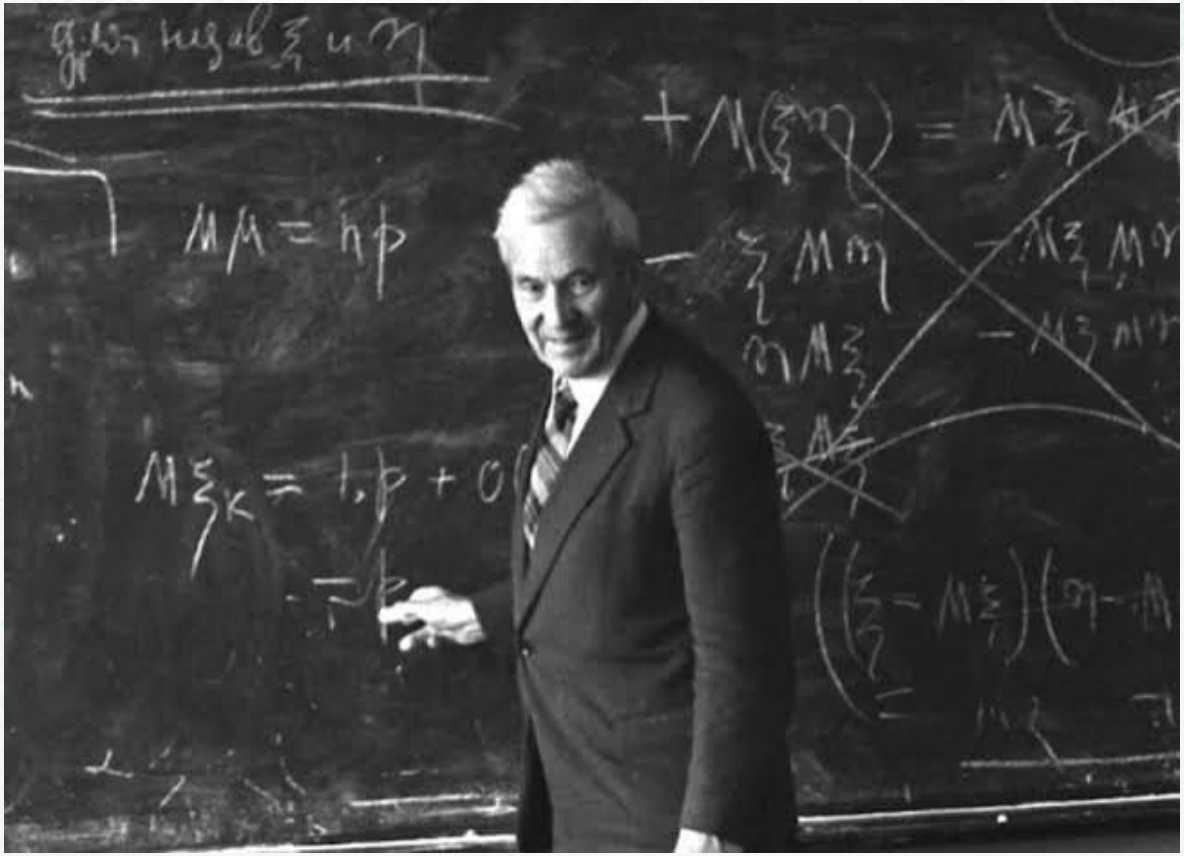
A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH AND WALTER PITTS

FROM THE UNIVERSITY OF ILLINOIS, COLLEGE OF MEDICINE,
DEPARTMENT OF PSYCHIATRY AT THE ILLINOIS NEUROPSYCHIATRIC INSTITUTE,
AND THE UNIVERSITY OF CHICAGO

Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

Perché comprensione ≅ compressione





Grazie, intelligenze naturali!

Se vi è piaciuto il talk
(ma anche se non vi è piaciuto):

- iscrivetevi al mio canale YouTube
- leggete i miei libri
- leggete i miei articoli su *Prisma magazine*
- invitatemi a parlare o, almeno, offritemi da bere.